

BFF10

Book of Abstracts

10th Bayesian, Fiducial, and Frequentist Conference
Salzburg, Austria

Compiled from speaker submissions · organized by session, in presenting order

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Keynote 1

E-Values and BFF: they should be BFF!

Peter Grünwald

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E-values are an alternative to p-values, providing a notion of statistical evidence that is simultaneously more robust and flexible. Here I will review and study e-values from the BFF angle. While e-values are frequentist “at heart”, their construction often involves the use of priors. I will extensively compare such e-values to Bayesian methods, contrasting (a) e-value and Bayes factor-based tests, (b) showing how e-based (anytime-valid) confidence intervals get wide rather than wrong if the prior is badly chosen; (c) showing how very Bayes-like quantities arise even for ‘nonparametric’ e-values that superficially do not look like Bayes factors or even likelihood ratios at all. Finally I turn to e-based decision theory, where the “e-posterior” interpolates between a standard Bayes posterior and Martin’s inferential models, thereby establishing a firm connection between e-values and the fiducial tradition as well.

Keynote 2

Outrigger local polynomial regression

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Standard local polynomial estimators of a nonparametric regression function employ a weighted least squares loss function that is tailored to the setting of homoscedastic Gaussian errors. We introduce the outrigger local polynomial estimator, which is designed to achieve distributional adaptivity across different conditional error distributions. It modifies a standard local polynomial estimator by employing an estimate of the conditional score function of the errors and an ‘outrigger’ that draws on the data in a broader local window to stabilise the influence of the conditional score estimate. Subject to smoothness and moment conditions, and only requiring consistency of the conditional score estimate, we first establish that even under the least favourable settings for the outrigger estimator, the asymptotic ratio of the worst-case local risks of the two estimators is at most 1, with equality if and only if the conditional error distribution is Gaussian. Moreover, we prove that the outrigger estimator is minimax optimal over Hölder classes up to a multiplicative factor $A_{\beta,d}$, depending only on the smoothness $\beta \in (0, \infty)$ of the regression function and the dimension d of the covariates. When $\beta \in (0, 1]$, we find that $A_{\beta,d} \leq 1.69$, with the limit of $A_{\beta,d}$ equal to 1 as β decreases to 0. A further attraction of our proposal is that we do not require structural assumptions such as independence of errors and covariates, or symmetry of the conditional error distribution. Numerical results on simulated and real data validate our theoretical findings; our methodology is implemented in R and available on CRAN.

Session 1

Graph Release with Assured Node Differential Privacy

Tianxi Li

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Differential privacy is a well-established framework for safeguarding sensitive information in data. While extensively applied across various domains, its application to network data — particularly at the node level — remains underexplored. Existing methods for node-level privacy either focus exclusively on query-based approaches, which restrict output to pre-specified network statistics, or fail to preserve key structural properties of the network. In this talk, I will introduce GRAND (Graph Release with Assured Node Differential privacy), which is, to the best of our knowledge, the first network release mechanism that releases entire networks while ensuring node-level differential privacy and preserving structural properties. Under a broad class of latent space models, we show that the released network asymptotically follows the same distribution as the original network. The effectiveness of the approach is evaluated through extensive experiments on both synthetic and real-world datasets.

Inferring model structure from data: MCMC for DAGs

Radu Craiu

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Inferring a directed acyclic graph (DAG) given data is computationally challenging. Current state-of-the-art MCMC methods for graph inference efficiently scan the space by first considering a restricted search space and iteratively expand the space until a stopping criterion is met. We estimate the error introduced from current methods that use restricted spaces compared to the full space and develop a novel MCMC method that reduces this error. The new adaptive algorithm allows for either expansion or contraction of the search space at any iteration. Extensive numerical experiments demonstrate the efficiency of the new algorithm and compare its performance with existing methods. This is joint work with Kieran Campbell and Morris Greenberg.

Combinatorial Extrapolation

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Predicting the effects of unseen combinations from limited observations is a fundamental challenge in domains such as drug discovery and hyperparameter optimization. We study combinatorial extrapolation, where training data contain only axis-aligned samples with a single active covariate, while test inputs involve multiple active covariates. We introduce DExtrl, a method for extrapolating interaction effects beyond the support of the training data. We establish theoretical conditions under which such extrapolation is possible and demonstrate on synthetic and real-world datasets that DExtrl accurately predicts outcomes for previously unseen combinations. We conclude with an outlook on perturbation-based cellular models, where current approaches still leave considerable room for improvement in capturing higher-order interactions. This is joint work with Marin Sola, Xinwei Shen and Michael Vollenweider.

Session 2

The E-measure

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We introduce the E-measure: a measure-like generalization of the E-value to a class of hypotheses. Unlike classical measures, E-measures are closed under infimums instead of addition. They arise from a compatibility axiom with logical implications, that there should be at least as much evidence against more specific hypotheses. We show that E-measures are the only non-dominated such objects, if the hypothesis class is closed under intersections. We propose to use the E-measure to present all the relevant evidence for a problem, where the relevance is captured by the choice of hypothesis class. We showcase this by applying the E-measure to decision making, inducing a hypothesis class from the uncertain consequences of decisions. This results in uniform E-consequence bounds on decisions, which nest high-probability loss bounds. Correcting for multiplicity, we consider ‘familywise evidence’ and ‘false evidence rate’ control, generalizing from errors and discoveries to continuous evidence. Remarkably, E-measures control these without multiplicity correction if the hypothesis class is intersection-closed. Moreover, we obtain a ‘frequentist’ notion of updating from E-prior to E-posterior. Abstracting the notion of a ‘hypothesis’, we advocate for using E-measures for any unknown quantity, leading to predictive E-measures.

E-values for Contingency Tables

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Analyzing many contingency tables simultaneously is important in the context of genetic association studies when analyzing associations between categorical genetic markers and a categorical (often binary) disease status. The research question that we are interested in is how to define an e-value for a contingency table, which is easy to compute and powerful. By “easy to compute”, we mean that resource-intensive operations like a loop over all contingency tables with given marginal counts shall be avoided. This requirement refers to the situation that hundreds of thousands of such e-values have to be computed for one and the same dataset in the case of a genome-wide association study (GWAS). By “powerful”, we mean that the e-value should (with high probability) be larger than the p-value based on Fisher's exact test calibrated to the e-value scale by a p-to-e-calibrator. Our primarily intended use case is a GWAS which is either multi-centric (independent patient groups are recruited at different locations) or group-sequential (independent patient groups are recruited at the same location at different time points). These two sampling schemes are realistic for GWAS, and e-values (which are easy to compute and powerful) can allow the data analysts to combine the evidence across centers or across time points, respectively, in a convenient manner (e.g., by multiplication or by averaging of e-values). For the special case of a 2×2 contingency table, we discuss several possibilities to define such e-values, and we characterize sampling schemes under which the usage of each of these e-values is particularly appropriate. This is joint work with Yongqi Wang, Sebastian Arnold, Francesca Giuffrida, and Peter Grünwald.

Multivariate Tail Probability Approximations

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This manuscript extends the univariate Lugannani and Rice saddlepoint tail approximation to multiple dimensions. The resulting approximation uses a multivariate Gaussian approximation to the distribution of the signed roots of the log likelihood statistics. As in the univariate case, the next correction term involves the difference between reciprocals of the signed root of the likelihood statistics and the analogous Wald statistics. Extensions to lattice variables and conditional distributions are provided.

Note: Thanks for the invitation.

Session 3

Uncertainty quantification via cross-validation and its variants under algorithmic stability

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Recently, there has been substantial interest in statistical guarantees for cross-validation (CV) methods of uncertainty quantification in statistical learning (cf. Barber et al. (2021a), Liang and Barber (2025), Steinberger and Leeb (2023)). These guarantees should hold under minimal assumptions on the data generating process and conditional on the training data, because numerous predictions are usually computed based on one and the same training sample. We push this objective to the limit: We prove asymptotic conditional conservativeness of CV, that is, the probability of the actual coverage probability, conditional on the training data, undershooting its nominal level vanishes asymptotically, under minimal assumptions. In particular, we impose a stability condition, require that the prediction error is stochastically bounded, and show that neither condition can be dropped in general. By way of an asymptotic equivalence result, we also show that the closely related CV+ method of Barber et al. (2021a) provides exactly the same conditional statistical guarantees as CV in large samples, thereby extending the range of applicability of CV+ to the high-dimensional regime. We conclude that, in view of its marginal coverage guarantee, CV+ does indeed improve over simple CV. For our proofs, we introduce a new concept called Lévy gauge, which can be of independent interest.

The Implicit Bootstrap

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Inference in many complex models is challenging due to analytical and numerical obstacles, often arising from the presence of latent variables introduced in a core model to account for data features such as truncation, censoring, other types of missing (not at random) data, misclassification, complex dependencies, and others. In these instances, deriving consistent estimators, which might require, e.g., methods for approximating integrals, is not only challenging but can be prohibitive when using bootstrap methods for inference. Moreover, numerical approximations might introduce finite sample bias, which can jeopardize the inferential accuracy of methods based on them. We propose an alternative bootstrap-type percentile method, namely the Implicit Bootstrap (IB), that bypasses the need for the specification (and computation) of consistent estimators for models incorporating complex data features. It is based on auxiliary statistics that can be defined implicitly but are computationally easier to obtain than consistent estimators. Typically, these auxiliary statistics are estimators associated with the core model, i.e., ignoring the data features, and the latter are introduced in the inferential procedure using simulations. We demonstrate the first and second order properties of the IB, under similar requirements used to establish the second-order accuracy of the studentized bootstrap, without the necessity of a consistent estimator. We present empirical evidence supporting the theoretical claims through simulation studies involving challenging scenarios.

Uncertainty quantification for functionals in constrained inverse problems

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Ill-posed inverse problems are characterized by a large set of parameters that are consistent with the observed data. Rigorous and useful finite-sample inference in these problems is possible by inferring functionals of the unknown parameter subject to physical constraints. The presence of the functional, constraints and potential unidentifiability makes this a challenging inference problem. In this talk, I will show that confidence intervals in this setting can be obtained through the inversion of a specific, non-standard likelihood-ratio test. The critical values of this test are typically not available in closed form so we propose calibrating the test over a bounded subset of the parameter space chosen so that it contains the true parameter with a high probability. I will then describe concrete computational strategies based on optimization and sampling for performing this calibration and test inversion in practice. I will first illustrate these ideas with low-dimensional toy examples and then present a realistic application to a moderately high-dimensional particle unfolding problem, where the new confidence intervals are significantly shorter than previous alternatives while maintaining correct finite-sample coverage. Time allowing, I will discuss the potential for carrying out this type of inference in ultra-high-dimensional data assimilation problems as well as recent extensions to the case of multiple functionals of interest.

Session 4

Addressing Small-Sample Challenges in Rare Disease Research Using Nonparametric Methods

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Rare diseases present unique challenges for the statistical analysis. Relevant outcomes often need to capture multiple aspects of the disease and may include quantitative, validated, and patient-relevant endpoints. In addition, sample sizes are typically small, and missing data can further reduce the available information. Statistical methods therefore need to account appropriately for these characteristics. To address these challenges, we systematically evaluate a range of univariate and multivariate nonparametric methods and assess their empirical performance. Although no universal solution emerges, the developed simulation framework provides detailed insights into the scope of applications of the different approaches. These findings may support methodological recommendations grounded in both statistical evidence and clinical expert opinion.

Note: Requested talk to be scheduled for Friday, July 10.

Two-Layered Logics for Uncertainty

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Reasoning about information, its potential incompleteness, uncertainty, and even contradictoriness need to be dealt with adequately. Various logical systems for reasoning in contexts when the available data about observed events are uncertain, unknown, or unreliable, have been considered. They use varied syntactical frameworks to capture semantical aspects like degrees of truth, likelihood, or subjective belief. Separately, incompleteness, uncertainty, and contradictoriness of information have been captured successfully by various logical formalisms and (classical) probability theory. While incompleteness and uncertainty are typically easily accommodated within one formalism, e.g. within various models of imprecise probability, contradictoriness and uncertainty less so — conflict or contradictoriness of information is rather chosen to be resolved before entering the reasoning process than to be reasoned with. One way to reason with conflicting information is to separate positive and negative support — evidence in favour and evidence against a statement — in the semantics and quantify it rather independently. This two-dimensionality gives rise to propositional logics with bi-lateral semantics: they are interpreted over twist-product algebras, bi-lateral frame semantics or bi-lattices, the well known Belnap-Dunn logic of First Degree Entailment being a prominent example. Belnap-Dunn logic with its double-valuation frame semantics can in turn be taken as a base logic for defining various uncertainty measures like probabilities on de Morgan algebras or belief functions. Two-layered logics, a simple two-sorted format developed for reasoning about probabilities of classical events, separate two layers of reasoning: the inner layer consists of a logic chosen to reason about events or evidence, the connecting modalities are interpreted by a chosen uncertainty measure on propositions of the inner layer, typically a probability or a belief function, and the outer layer consists of a logical framework to reason about probabilities or beliefs. Logics originally introduced by Fagin, Halpern, and Megiddo use classical propositional logic on the inner layer to capture events, a probability measure to interpret a modality, and systems of linear inequalities on the outer layer. Hájek, Godo, and Esteva, on the other hand, suggested using many-valued Łukasiewicz logic on the outer layer, to capture the quantitative reasoning about probabilities within a propositional logical language. Our main objective is to extend the apparatus of two-layered modal logics for the formalisation of reasoning with uncertain information, which itself might be non-classical, i.e., incomplete or contradictory. Many-valued logics with a bi-lateral semantics can be used on the outer layer to reason about belief, likelihood or certainty based on potentially incomplete or contradictory evidence, building on Belnap-Dunn logic of First Degree Entailment as an inner logic of the underlying evidence. This results in two-layered logics suitable for various scenarios: expansions of Łukasiewicz logic are adequate in cases when (possibly aggregated) evidence yields a pair of probability measures or a belief function (on a De Morgan algebra), while expansions of Gödel logic are useful to reason about comparative uncertainty, or to capture reasoning about qualitative probability.

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Uncertainty Propagation in System Reliability: From Inference on Components to System Risk

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Reliability is defined as the probability that a system provides its desired functionality — it is the probability of an event. Applications to risk analysis commonly drive decisions about singular trials with no relation to repeated sampling grounding utility benefits by long-term averages. Such problems are specific by dealing with high stakes (catastrophic failures, loss of lives) and insufficient data (small sample size, interval censoring, missing data). Fully probabilistic models tend to include unaccounted assumptions which could lead to false confidence. Imprecise probability theories emerged to rectify this by embracing ignorance into the mathematical models, leading to more conservative but more trustworthy assessments. Expensive complex systems may be ill-suited for subjection to destructive testing for the sake of statistical analysis. Systems reliability theory decomposes the failure event into sub-events associated with functionalities of subsystems and components, and the original system failure event is subsequently represented with a function dependent on the components' state. Statistical inference is usually conducted for the individual components and needs to be propagated to obtain an integrated assessment about the whole system. This propagation poses challenges both theoretically (to ensure validity of the top-level assessments) and numerically (especially for Imprecise Probabilistic models, which include extremization subproblems). In the talk, we will discuss theoretical and practical aspects of imprecise reliability models, propagation of component-level inferential results, and applications to risk-based decision making.

Session 5: Special Invited Session

Celebrating the 10th BFF Conference

From BFF10 to Quantum Probability 100 (and back): Can Broadening Probability Deepen Statistics?

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Perhaps echoing the saying that “Coincidence is God’s way of remaining anonymous,” the COVID-delayed celebration of BFF10 coincides with the centenary of quantum probability, commonly traced to Max Born’s 1926 probabilistic interpretation of the wave function, whose temporal propagation is described by the Schrödinger equation, also published in 1926. Despite this long history, quantum probability remains unfamiliar, even intimidating, to many statisticians. Motivated by the coming era of quantum computing and by foundational questions surrounding the Bayesian, fiducial, and frequentist (BFF) schools, this talk explains the contextual probability framework developed by physicist and philosopher Jacob Barandes to better connect quantum theory with ordinary notions of probability. By relaxing the commitment to a common probability space, Barandes’ framework does not require every probability distribution to arise from a single underlying (possibly multivariate) random variable, but instead relates distributions across different, sometimes incompatible, contexts. In this setting, superposition, interference, and the nonexistence of joint distributions arise naturally while preserving a distinction between mathematical representation and physical reality. The talk then observes that analogous phenomena already exist within classical statistics, ranging from improper priors and incompatible conditional distributions to fiducial representation non-invariance, and invites the audience to contemplate whether these similarities are merely suggestive coincidences or hints of a broader probabilistic framework for comparing, contrasting, and enriching the foundations of the BFF schools.

Session 6

Sequential Bayesian experimental design for prediction in physical experiments informed by computer models

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In many scientific and engineering domains, physical experiments are often costly, non-replicable, or time-consuming. The Kennedy & O'Hagan (KOH) model framework has become a widely used approach for combining simulator runs with limited experimental observations. Under a Bayesian implementation, the simulator output, model discrepancy, and observation noise are jointly modeled by coupled Gaussian processes, followed by coherent posterior inference and uncertainty quantification. This work presents a genuinely sequential Bayesian experimental design (BED) framework explicitly aimed at improving the predictive performance of the KOH model. We employ a mutual information (MI)-based criterion and develop a hybrid variant that integrates with measures of local model complexity, leading to significantly more efficient design decisions. We further theoretically establish asymptotic links between the MI-based criterion and the classical integrated mean squared prediction error (IMSPE) minimization criterion. In practice, we find the MI-based criterion is more comprehensive and robust than the IMSPE minimization criterion, especially when the model is highly uncertain in the early stages of the experiment. We demonstrate the effectiveness of the proposed methods through both a synthetic example and a real biochemical case study, and compare the MI- and IMSPE-based criteria against several other classical design criteria under sequential (offline) and adaptive (online) BED settings.

Towards the next generation of multi-analyst experiments

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Data analysis is a linchpin of modern scientific research. However, the impact of the data analysis pipeline on the reliability of inferences has been understudied. So-called multi-analyst experiments have begun to address this deficit, resulting in claims that the data analysis pipeline itself may be a primary source of unreliability. Unfortunately, due to the research designs employed, the interpretability of those results is limited. In particular, those designs do not allow for the identification of the sources of unreliability. Hence, it is difficult to use their results to constructively identify ways to improve data analysis practices. We attempt to identify and address these limitations with a new design-of-experiments framework, thereby enabling future multi-analyst studies to draw more specific and actionable conclusions.

Understanding prediction Risk in Large Language Models

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Prediction risk analysis in large language models is of great importance for the understanding of trustworthiness of AI tools, in which uncertainty quantification plays a central role. Using a token-level foundational decomposition for the prediction risk, we present a unified framework involving Variability, Confidence and Innateness (VCI) to depict different types of uncertainty. Such decomposition is also extended to agentic systems for prediction risk analysis. The replicability for scientific research is used as a running example to demonstrate the decomposition for risk analysis and uncertainty quantification. This is a joint work with Jiuchen Zhang and Rui Nie.

On Theoretical Foundations of Extended Fiducial Inference

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Extended fiducial inference (EFI) provides a general paradigm for statistical inference in modern data science. Its defining feature is to formulate statistical inference as a problem of solving data-generating equations under uncertainty: EFI first constructs a fiducial law for the unobserved random errors underlying the observations and then propagates this law to the parameter space, either through an inverse mapping or, more generally, through a compatibility measure defined on the inverse fiber. This formulation enables EFI to address several difficulties that arose in Fisher's original fiducial argument, including issues related to conditioning, marginalization, and noninvertibility. We develop general theoretical foundations for EFI that encompass both single-valued and set-valued inverse mappings and apply to both low- and high-dimensional regimes. We illustrate the theory through linear regression, nonlinear regression, logistic regression, their high-dimensional counterparts, and other analytical examples.

Session 7

Universal ordinal inference with extended ranks

Peter Hoff

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The predictive accuracy of a regression model depends largely on the distributional support of the outcome variable and the relationship between the mean and variance of the conditional distribution. In this article we develop a single inferential framework that can represent these aspects for a wide range of conditional distributions. Our approach is built upon a monotonically transformed linear regression model and a pseudo-likelihood based on an extended notion of ranks. This extended rank likelihood approach can accommodate any ordinal data type, including continuous and discrete ordered data, and can represent a wide range of mean-variance relationships. Theoretical results show that for parameter estimation, the maximizer of this likelihood is asymptotically efficient at the two extremes of ordinal data — continuous and binary data. Bayesian parameter estimates and posterior prediction intervals are available via a simple Gibbs sampling algorithm. Prediction intervals with guaranteed frequentist coverage are constructed via conformal calibration of the Bayesian posterior distribution.

Bayesian nonparametric spectral analysis of locally stationary processes

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Many real-world phenomena can well be modelled by locally stationary time series with a slowly changing dependency structure. For such time series the estimation of the time-varying spectral density is of particular interest. In this talk we propose a Bayesian nonparametric method for this purpose based on a suitable dynamic Whittle likelihood for locally stationary time series in combination with a Bernstein-Dirichlet process prior. We prove sup-norm posterior consistency and obtain L2-norm posterior contraction rates. Additionally, this methodology enables model selection between stationarity and non-stationarity based on the Bayes factor. The good finite-sample performance of the method is indicated by means of a simulation study and applications to real-life data-sets. This is joint work with Yifu Tang (University of Otago), Kate Lee and Renate Meyer, both from the University of Auckland.

Bayesianism, Frequentism, and Imprecise Probabilities

William Peden

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To what extent can imprecise probabilities help us combine the strengths of both Bayesian and frequentist approaches? This paper describes a 5-year collaborative research programme that helps to answer this question, conducted by an interdisciplinary team from philosophy, decision theory, and machine learning. The differences between Bayesianism and frequentism are less important when we have rich information about the data-generating process. Instead, the most interesting differences between Bayesianism and frequentism occur under ambiguity: when there is a lack of information about the true probabilities. To make comparisons under ambiguity, we used a tractable decision problem, the “Bernoulli problem”, based on sequences of Bernoulli trials. Agents know that the trials are exchangeable and binomial, but they do not know the true probabilities or even bounds on those probabilities. We investigated imprecise probability learning systems, which can combine Bayesian and frequentist statistical methods. One popular such system is Imprecise Bayesianism, which uses sets of probability distributions. Its key strength is that it can represent the ambiguity in a decision problem through divergence among the probability distributions in an agent’s set. We found a surprising trade-off that we call the Ambiguity Dilemma. The tools for representing ambiguity have a systematic performance cost, compared to traditional Bayesianism, in the Bernoulli problem. Despite the reliable information that is available even in small samples of Bernoulli trials, an Imprecise Bayesian with divergent sets of beta distributions learns slowly, resulting in inferior decision-making performance in this problem. We subsequently found that the Ambiguity Dilemma is very robust. For instance, it also occurs for Dempster-Shafer updating. However, the Ambiguity Dilemma is avoided by Calibration imprecise probability systems, which combine aspects of frequentism and Bayesianism. A Calibration agent can retain the ambiguity-representation tools of Imprecise Bayesianism while achieving learning performance comparable to the best traditional Bayesian agents in the Bernoulli problem. Thus, Calibration imprecise probability systems exemplify the value of integrating Bayesian and frequentist methodological ideas. Our current focus is on comparisons in more complex and realistic decision problems, such as those with noisy data.

Session 8

A Plea for History and Philosophy of Statistics and Machine Learning

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The integration of the history and philosophy of statistics was initiated at least by Hacking (1975) and advanced by Hacking (1990), Mayo (1996) and Zabell (2005), but it has not received sustained follow-up. Yet such integration is more urgent than ever, as the recent success of artificial intelligence has been driven largely by machine learning—a field historically developed alongside statistics. Today, the boundary between statistics and machine learning is increasingly blurred. What we now need is integration, twice over: of history and philosophy, and of two fields they engage—statistics and machine learning. I present a case study of a philosophical idea in machine learning theory (and in formal epistemology) whose root can be traced back to an often under-appreciated insight in Neyman and Pearson’s 1936 work on uniformly most powerful unbiased tests (a follow-up to their 1933 classic). This leads to the articulation of an epistemological principle—largely implicit in, but shared by, the practices of frequentist statistics and machine learning—which I call *achievabilism*: the thesis that the correct standard for assessing non-deductive inference methods should not be fixed, but should instead be sensitive to what is (provably) achievable in specific problem contexts. I close by putting forward the conjecture that *achievabilism* also underlies the development of nonparametric Bayesian statistics by Freedman and Diaconis. If so, *achievabilism* is an epistemological principle shared by many frequentists, Bayesians, and machine learning theorists.

Note: Thanks for organizing the conference :)

The method of arbitrary functions and frequentist probabilities

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One important criticism of the frequentist interpretation of probability is that it is logically circular: probability cannot be defined as the limiting frequency of a random sequence via the law of large numbers because the law of large numbers depends on the probability law of the random variables in the sequence. In this talk, I explore how Poincaré's and Hopf's method of arbitrary functions may be used to define probabilities as frequentist limits of deterministic dynamical systems that forget initial conditions. This idea may also help explain how LLMs produce outputs with stable distributions despite variations in prompt inputs.

Some confusing issues with frequentism that a Bayesian or fiducial approach cannot solve

Christian Hennig

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I will present three rather simple but problematic issues with frequentism, namely (a) the misspecification paradox (checking and accepting a model assumption by a misspecification test actively violates the model), (b) the impossibility to identify many elementary dependence patterns from data (such as constant correlation between Gaussian observations), (c) an estimator that is arguably bad for large n because of consistency, namely the BIC used for estimating the number of clusters in a mixture model. I will then discuss how these issues are not exclusively problems for frequentism, but rather have to do with our use and understanding of probability models more generally (particularly requiring them to be “true” in some sense).

Session 9

Confidence Intervals for Rate Estimation with Importance Sampling in Autonomous Vehicle Evaluation

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Accounting for both rare events and complex sampling presents challenges when quantifying uncertainty for rate estimation in autonomous vehicle performance evaluation. In this paper, we introduce a statistical formulation of this problem and develop a unified compound Poisson model framework for unbiased rate estimation through the Horvitz-Thompson estimator. Though asymptotic theory for the model is available, the inference of confidence intervals (CIs) in the presence of rare events requires new investigation. We also advocate for a new monotonicity criterion for rate CIs — summing the rates of disjoint types of events should produce not only a higher point estimate but also higher confidence bounds than for the individual rates — that facilitates interpretability in real applications. We propose a novel exponential bootstrap (EB) method for CI construction based on a fiducial argument; it satisfies the monotonicity property, while novel extensions of some existing methods do not. Comprehensive numerical studies show that EB performs well for a wide range of settings relevant to our applications. Fast implementation of EB based on saddlepoint approximation is also developed, which may be of independent interest.

Advancing Decision-Efficiency in (Pre)-clinical Research via Nonparametric Sequential Frameworks

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Traditional sequential designs in early-stage (pre)-clinical research typically rely on parametric assumptions that are frequently violated by small sample sizes, heavy tails, or extreme outliers. To address these limitations and optimize decision efficiency, we introduce a novel, purely nonparametric sequential framework based on exact and asymptotic rank procedures. This distribution-free framework integrates multi-stage interim analyses with sequential rank-sum statistics, allowing early termination for efficacy or futility without assuming underlying normality. We evaluated the framework's performance using extensive Monte Carlo simulations, benchmarking it against traditional fixed-sample and classical parametric group sequential designs across various non-normal and highly skewed data distributions. The proposed rank-based framework significantly reduced the expected sample size. Crucially, it maintained robust Type I error control and preserved target statistical power where parametric alternatives failed due to model misspecification. By providing a mathematically rigorous, distribution-free approach to real-time data monitoring, this framework offers a highly ethical and resource-efficient paradigm for accelerating go/no-go decisions in modern (pre)-clinical translation.

Fairness Properties for likelihood ratios in the interpretation of forensic evidence

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In the construction of values of forensic evidence, formal methods associating a piece of (trace) evidence from an unknown source to a specified source typically result in Bayes factors. They are constructed by assuming formal priors for the nuisance parameters of the specified source's probability distribution of traces, and the nuisance parameters of the relevant background population of sources' probability distribution of traces. These prior distributions may vary in the degree of information they contain about these nuisance parameters. Since we normally have limited measurements on sources and a limited number of control samples, the following becomes important: we need to pool information across sources; if not, Bayes factors become highly dependent on the choice of prior distributions, due to limited information to balance out the prior information. Our recent work studies the relation of the information present in these prior distributions, and the pooling choices, to biases naturally arising in Bayes factors and also in score-based likelihood ratios. These biases may either lead to negative discrimination of rare suspects, or to positive discrimination of rare suspects. In this presentation, we will define what we see as a fair use of evidence; show what type of priors (related to types of Bayes factors found in the forensic literature) lead to what type of bias; show how pooling for score-based likelihood-ratio leads to bias; and discuss score-based strategies for reducing the dimension of the parameter space and strategies for mitigating the bias against the specified source that arises from pooling information in a score-based approximation to fair values of evidence.

Note: Joint work with Dr. Peter Vergeer of the Netherlands Forensic Institute.